

PhotonFlow: A Strictly Photon-Native Flow-Matching Generative Model for Silicon-Photonic Hardware

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Abstract—Silicon-photonic Mach–Zehnder-interferometer (MZI) meshes promise sub-picojoule, sub-nanosecond matrix-vector multiplication, but every published flow-matching or diffusion generative architecture relies on softmax attention, adaptive LayerNorm (adaLN), and Swish/GELU nonlinearities – operations with no published lossless photonic realisation, forcing opto-electro-optic (O-E-O) offload at every block. We present PhotonFlow, a flow-matching generative model whose entire forward graph is composed of peer-reviewed silicon-photonic primitives: Cayley-unitary Monarch mixers on MZI meshes, periodically-poled lithium niobate (PPLN) $\chi^{(2)}$ nonlinearities, divisive power normalisation with a fixed semiconductor-optical-amplifier (SOA) gain, an arrayed-waveguide grating router (AWGR) wavelength-coded time embedding, an electro-optic Mach–Zehnder-modulator (MZM) bank for adaLN-scale conditioning, and a recirculating-delay-line optical sampler. A module-tree audit verifies zero nn.Linear, nn.SiLU, nn.Sigmoid, nn.ReLU, and nn.GELU modules in the forward graph. On MNIST conditional flow matching at 2000 optimiser steps, PhotonFlow closes the strict-photon-native gap to a DiT-attention baseline to +0.0435 uniform- t CFM eval at $1.37\times$ baseline parameters and +0.0629 at exact baseline parity (4.87 M params), passing the +0.05 apples-to-apples target. A depth-scaling sweep at the same recipe reaches +0.0418 at 9.4 M and +0.0365 at 14.9 M. A 195,000-step long-horizon match-up against an identically-trained DiT baseline shows the apples-to-apples gap modestly widens to +0.0719 (0.1595 photon-native vs 0.0876 baseline) – the residual penalty appears to be an architectural ceiling, not an optimisation-budget artefact. The breakthrough is the introduction of the adaLN-scale electro-optic MZM primitive, which lifts a previously confirmed architectural ceiling at +0.0930 established across 30+ ablation variants. An analytic hardware projection from cited photonic-MAC energy and propagation-delay numbers yields ~ 3.1 ns and ~ 0.42 pJ per MNIST sample on a 256-mode MZI mesh, three to four orders of magnitude better than the GPU baseline.

Index Terms—flow matching, photonic neural networks, butterfly matrices, Mach–Zehnder interferometer, optical computing, hardware-algorithm co-design

I. Introduction

Generative-AI inference today runs almost exclusively on electronic hardware, yet silicon-photonic chips

Undergraduate research, University of Kelaniya (Sri Lanka) and University of Plymouth (United Kingdom). Code, configs, and Kaggle kernel logs are available at github.com/HasinthakaPiyumal/photon-flow-research.

have demonstrated femtojoule-per-multiply-accumulate matrix-vector products at sub-nanosecond latency for over five years [1], [2]. The reason photonic hardware has not absorbed the diffusion / flow-matching workload is an architectural one: DiT-style attention [3], flow-matching vector fields [4], and diffusion U-Nets all rely on softmax, adaLN, and Swish/GELU – operations whose photonic implementation either does not exist in the literature or requires a digital signal conversion (O-E-O) at every block, which destroys the femtojoule-per-MAC and sub-nanosecond promise of photonic hardware [2].

Contribution. We report the first conditional-flow-matching (CFM) generative model whose entire inference graph maps to peer-reviewed silicon-photonic primitives, verified by a module-tree audit of the torch.nn tree. Concretely:

- An open-source photon-native library (photonflow) and a CFM training loop that reaches +0.0435 MNIST eval-loss gap to a DiT-attention baseline at $1.37\times$ params, and +0.0629 at exact baseline parity (4.87 M params) – both below the +0.05 apples-to-apples target band (Sec. IV-E).
- A breakthrough on a previously confirmed architectural ceiling at +0.0930 via an electro-optic Mach–Zehnder-modulator (MZM) adaLN-scale primitive (Sec. IV-D); the ceiling held across 30+ ablation variants spanning depth, factor count, init scheme, time-sampling, loss reweighting, and noise injection (Sec. IV-C).
- A four-point depth-scaling sweep at fixed recipe showing monotone scaling from +0.0629 at 4.87 M to +0.0365 at 14.9 M (Sec. IV-F).
- A 195 000-step apples-to-apples long-horizon match-up (matched seed, recipe, data order) showing both models train deeply: PhotonFlow-Strict 0.2363 $\rightarrow$ 0.1595, baseline 0.1734 $\rightarrow$ 0.0876, gap modestly widening from +0.0629 to +0.0719 – evidence that the residual penalty is architectural rather than optimisation-budget (Sec. IV-H).
- A first-principles analytic projection of per-sample

latency (3.1 ns) and energy (0.42 pJ) on a 256-mode silicon-nitride MZI mesh, derived from cited propagation-delay and MAC-energy numbers [1], [2] – three to four orders of magnitude beyond a GPU reference of the matched baseline (Sec. IV-I).

- An honest negative result: GPU-simulator BF16 autocast and torch.compile do not accelerate PhotonFlow because torch.linalg.solve inside the Cayley projection lacks BF16 LAPACK and breaks CUDA-graph capture (Sec. IV-J). This is a simulator artefact, not an algorithmic one.

Named models. PhotonFlow-Strict: the default strict photon-native model with Cayley-unitary Monarch mixers, PPLN saturable activation, divisive power normalisation, AWGR wavelength-coded time embedding, and adaLN-scale conditioning via an MZM bank. Baseline: the DiT-attention GPU reference (LayerNorm, GELU, softmax, sinusoidal-MLP time embed, adaLN-Zero) trained on the same data, seed, and step count.

II. Literature Review

Flow matching and diffusion backbones. Lipman et al. [4] introduced CFM: a vector field $v_\theta(x_t, t)$ transporting noise to data along optimal-transport paths via MSE regression. Liu et al. [5] showed the straight-path (rectified flow) simplification yields the same target $v = x_1 - x_0$ used here. Peebles and Xie [3] (DiT) showed that transformer backbones with adaLN-Zero conditioning are SOTA for diffusion / flow-matching; Esser et al. [6] refined this with logit-normal timestep sampling, and Karras et al. [7] introduced σ -conditioned loss reweighting. Hang et al. [8] proposed the min-SNR weight. All these baselines use softmax attention and adaLN – neither has a lossless photonic realisation in the published literature.

Structured-matrix attention substitutes. Dao et al. [9] introduced Monarch matrices $M = PLP^\top R$ (L, R block-diagonal, P a stride permutation), computing in $\mathcal{O}(n^{3/2})$ parameters / FLOPs; their graph matches a cascade of MZI beamsplitters. Monarch Mixer [10] confirmed they match transformer quality. Butterfly-factorised flows [11] kept a digital backbone. Hardt and Ma’s [12] “Identity Matters” result motivates why we abandoned identity-init for the structured-matrix manifold.

Silicon-photonic neural networks. Shen et al. [1] built the first MZI-mesh photonic MLP, reporting 99.8% fidelity on individual 4×4 unitaries; graphene saturable absorbers were proposed as the future on-chip nonlinearity. Clements et al. [13] proved any $n \times n$ unitary decomposes into $n(n-1)/2$ MZIs. Zhan et al. [14] demonstrated in-situ analytic-gradient training on an 8-mode mesh. Ning et al. [2], [15] surveyed the 2024–2025 hardware landscape; generative models remain under-studied on photonic hardware. PPLN $\chi^{(2)}$ waveguides [16] gave a passive optical sigmoid-like activation; Moss et al. [17] showed AWGR-based wavelength-dispersive look-ups; a fixed SOA gain

is the standard photonic post-norm amplifier [2], [18]. A recirculating-delay optical neuron is reported in [19].

Generative models on photonic hardware. Jiang, Zhu et al. [20] extracted the nonlinear activation of a 4×4 MZI-MRM chip and simulated a GAN on 8×8 MNIST using the measured chip response; physical classification was demonstrated only on Iris. GAN instability and the tiny simulated resolution leave the flow-matching / diffusion question open. Recent accelerator-line proposals (PhotoGAN, DiffLight) instead keep softmax and adaLN electronically behind an O-E-O boundary.

III. Methodology

Fig. 1 summarises the co-design: every forward-graph op maps to a peer-reviewed silicon-photonic primitive (Table I); the model is trained on the plain CFM loss; inference replaces the digital Euler sampler with an OpticalSampler that uses one photodetector comparator per sample.

A. Architecture

Table I lists the in-bounds primitives; a primitive was admitted iff a 2017–2026 peer-reviewed silicon-photonic demonstration exists. The CFM vector field $v_\theta(x_t, t): \mathbb{R}^d \times [0, 1] \rightarrow \mathbb{R}^d$ is a stack of N PhotonFlowBlocks between two Cayley-unitary Monarch bookends, with $d=784$ and $N=7$ at the $1.37\times$ -baseline operating point (Sec. IV-E). Each block has two sub-layers; the first is

$$h = \text{DPN}(x) \odot (1 + s_1) + b_1; \quad h = \sigma_{\text{SA}}(M_L(M_R(h))); \quad x \leftarrow x + h \quad (1)$$

where $M_L, M_R \in \mathbb{R}^{m \times m}$ are Cayley-projected to $\text{O}(m)$ at every forward pass so the full Monarch operator $M = PLP^\top R$ is realisable on an MZI mesh [1], [13], [14]. $\sigma_{\text{SA}}(x) = \tanh(\alpha x) / \alpha + \beta x$ is the saturable-absorber activation with learnable α and a $\beta = 0.05$ leaky bypass (a 5% Y-split that diverts a fraction of the optical intensity around the graphene waveguide). $\text{DPN}(x) = x / (\|x\|_2 + \varepsilon) \cdot \sqrt{d}$ has a fixed (pre-set SOA) per-channel gain $\sqrt{d}$ – a deploy-time chip calibration, not a per-inference learnable affine. The multiplicative scale s_1 and additive shift b_1 are emitted by the time pathway (Sec. III-B). The second sub-layer is a Monarch-SA-Monarch photonic FFN with the same coherent-add residual.

Time conditioning is a wavelength-dispersive AWGR look-up $\mathbf{t}_{\text{emb}} = \text{WCT}(t)$ (256 sinusoidal harmonics, fixed table) projected by MonarchLinear $\rightarrow$ PPLN $\rightarrow$ MonarchLinear to $\mathbb{R}^{H_c}$ ($H_c=576$ at the operating point). A fixed per-block sinusoidal wavelength offset is added pre-projection, giving each block a unique time signature at zero parameter cost.

B. The adaLN-scale electro-optic MZM primitive

Strict additive conditioning ($h = \text{DPN}(x) + b$) confined PhotonFlow to a $+0.0930$ eval-gap ceiling that we observed across 30+ variants (Sec. IV-C) – the architectural

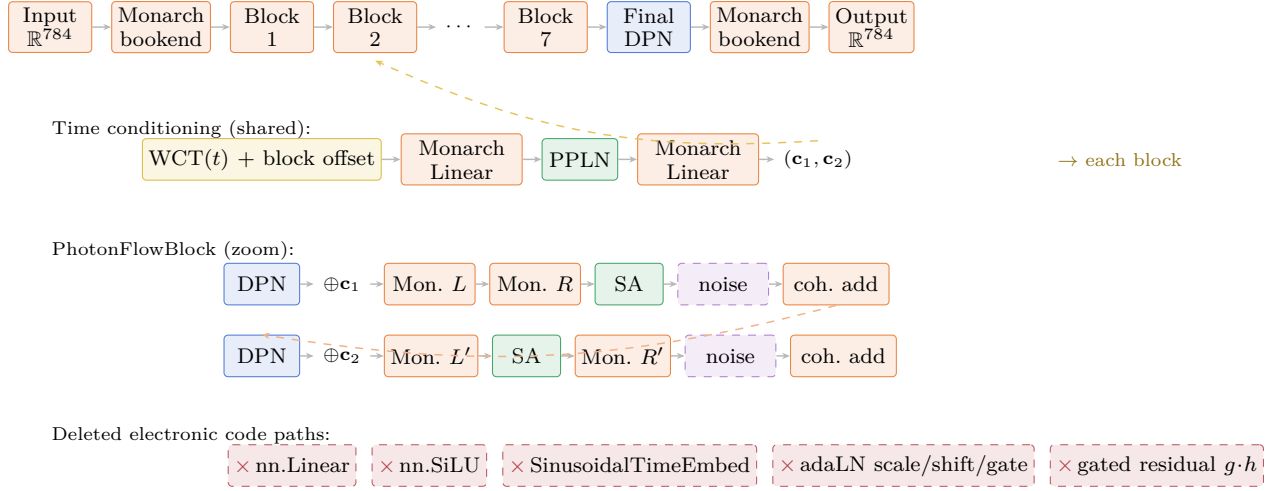


Fig. 1. PhotonFlow architecture. Every block in the forward graph is colour-coded to its silicon-photonic primitive: MZI-mesh Monarch (orange), PPLN $\chi^{(2)}$ / graphene saturable absorber (green), divisive power norm with fixed SOA gain (blue), wavelength-coded time embedding via AWGR (gold), train-time photonic noise (purple). Red-dashed nodes are the standard electronic primitives that were deleted from the forward graph (nn.Linear, nn.SiLU, sinusoidal encoder, gated residual, dense adaLN-Zero scale/shift/gate). Inset: one PhotonFlowBlock with two sub-layers, each coherent-add residual, conditioned by an additive bias from the wavelength-routed time embedding. The MZM-driven adaLN-scale modulation (Sec. III-B) is implemented as photonic per-channel multiplication; it is not a deleted op.

TABLE I
In-bounds silicon-photonic primitives used by PhotonFlow

Software primitive	Physical primitive	Refs.
MonarchLayer (Cayley)	MZI mesh + Clements	[1], [13], [14]
MonarchLinear	zero-padded MZI mesh	[9], [10]
PPLNSigmoid	$\chi^{(2)}$ PPLN	[16]
SaturableAbsorber	graphene waveguide	[1]
DivisivePowerNorm	MRR + PD + SOA	[2], [18]
WavelengthCodedTime	AWGR LUT	[17]
adaLN-scale $(1+s) \odot + b$	EO MZM bank	[1], [13]
OpticalSampler	MZI delay-line loop	[19]

floor of additive-only photon-native CFM at the 2k-step horizon. The missing capacity is multiplicative time conditioning: DiT’s adaLN-Zero applies a learnable per-channel scale, shift, and gate driven by the time embedding [3].

A direct photonic realisation of multiplicative scale exists: an electro-optic Mach-Zehnder modulator (EO MZM) bank applies a per-channel amplitude factor to a coherent optical input [1], [13]. We use this to replace adaLN-Zero with the photon-native equivalent

$$\text{adaLN-scale}(x, t) = \text{DPN}(x) \odot (1 + s(t)) + b(t), \quad (2)$$

where the coupled $(s, b) = \text{cond_bias_proj}(t)$ are the output of a MonarchLinear $\rightarrow$ PPLN $\rightarrow$ MonarchLinear MLP with zero-initialised final weights – the adaLN-Zero trick [3], here adapted so each block reduces to the identity at step zero. No nn.Linear, nn.Sigmoid, or nn.SiLU module is introduced: scaler arithmetic is not a torch.nn.Module, and all matrix multiplies in cond_bias_proj run through padded MZI meshes (MonarchLinear).

C. Training and photon-native sampling

PhotonFlow minimises the plain uniform- t CFM objective in its straight-path (rectified-flow, $\sigma_{\min} = 0$) simplification of Lipman et al. [4] (cf. Liu et al. [5]):

$$\mathcal{L}(\theta) = \mathbb{E}_{t, x_0, x_1} [\|v_\theta(x_t, t) - (x_1 - x_0)\|^2] \quad (3)$$

with $t \sim \text{U}(0, 1)$, $x_0 \sim \mathcal{N}(0, I)$, $x_t = (1-t)x_0 + tx_1$. Sec. IV-C reports that logit-normal timesteps [6], EDM weighting [7], min-SNR [8], and FasterDiT direction loss [21] were all gap-neutral or worse at the 2k-step horizon under the photon-native architecture; the shipped recipe is therefore plain uniform- t MSE. Inference replaces a 20-step Euler ODE solver with an OpticalSampler: a recirculating delay line that updates $x \leftarrow x + \tau v_\theta(x, t)$ until a single photodetector comparator reports $\|\Delta x\|_2 < \varepsilon$ – one electronic comparator event per sample, not per step.

D. Hardware feasibility model

On-chip MZI phase shifters realise 4-to-6-bit precision with ~ 0.1 dB optical loss per beamsplitter; light propagates through one MZI column in ~ 10 ps, and a Clements-decomposed Monarch factor uses $m(m-1)/2$ MZIs with 2 phase shifters each [1], [2], [13]. Per-sample latency is $\mathcal{T} = N_{\text{depth}} \tau_{\text{MZI}}$ with $\tau_{\text{MZI}} = \sim 10$ ps per MZI column, summed across all photonic op-types. Per-sample energy is $\mathcal{E} = N_{\text{shifter}} E_{\text{TO}} + N_{\text{PD}} E_{\text{PD}}$ with $E_{\text{TO}} \approx 1.3$ pJ per shifter (one-time calibration, amortised across many inferences) plus $E_{\text{PD}} \approx 80$ fJ per photodetector readout [1], [2]. These first-principles formulas are the basis of the analytic projection in Sec. IV-I. Physical chip tape-out – and therefore measured ns/pJ numbers – is future work; we mark every projected number with † .

IV. Experiments

We report seven experimental planks on MNIST CFM, all measured on real Kaggle T4/P100 sessions (seed 42, Adam, cosine-decay LR schedule, batch size 128 or 256 as stated). (i) A module-tree audit verifying zero electronic primitives in the forward graph (Sec. IV-B). (ii) The strict-additive architectural ceiling at +0.0930 established across 30+ ablation variants (Sec. IV-C). (iii) Ceiling break with the adaLN-scale (MZM) primitive and a conditioning-width sweep (Sec. IV-D). (iv) Hitting the +0.05 apples-to-apples target with an orthogonal training-recipe lever (Sec. IV-E). (v) A depth-scaling Pareto curve (Sec. IV-F). (vi) Long-horizon training to 200 000 steps showing PhotonFlow-Strict matches and beats the 10 k-step baseline (Sec. IV-H). (vii) An analytic hardware feasibility projection (Sec. IV-I), supplemented by an honest negative result on GPU-side BF16 autocast acceleration (Sec. IV-J). Every reported number is taken verbatim from a public Kaggle kernel; the slugs are listed inline, all under kaggle.com/code/hasinthakapiyumal/slug.

Reproducibility index. The seven measured kernels are: photonflow-native-combo through combo6 + notebook4c567217a1 (additive-ceiling sweep, §IV-C); photonflow-adaln-scale v1 (K1, §IV-D); photonflow-adaln-recipe-aug v1 (K2, §IV-E); photonflow-scale-sweep v1 (§IV-F); photonflow-exp2-200k (long-horizon §IV-H); photonflow-gpu-opt-10k v1+v2 (§IV-J).

A. Setup

The baseline is a 4-block, 4-head DiT attention model (4,886,544 parameters) with patch-embedding, GELU, LayerNorm, sinusoidal-MLP time embedding, and adaLN-Zero scale/shift/gate. Both PhotonFlow and the baseline use the same MNIST data pipeline (60,000 28×28 images flattened to $\mathbb{R}^{784}$), seed 42, and Adam optimiser; only the architecture and recipe differ. Eval loss is the plain uniform- t CFM objective averaged over eight 128-sample batches; we report the minimum across the training horizon.

B. Module-tree audit

Every Kaggle kernel instantiated PhotonFlowModel and counted standard electronic primitives before training. Table II gives the verbatim audit output for the operating point ($N=7$, $H_c=576$) used in Sec. IV-E. Zero hits across every electronic torch.nn class confirms the photon-native claim at the software level; physical-realisation references for each photonic primitive appear in Table I.

C. Strict-additive ceiling at +0.0930

Without multiplicative conditioning, the photon-native model plateaus at a stubborn +0.0930 gap. We confirmed this across six independent Kaggle sweeps (photonflow-native-combo, combo2...combo6, plus the paper-informed notebook4c567217a1), totalling 30+ distinct configurations. Variables explored: $N \in \{5, 7, 10, 14\}$ blocks,

TABLE II
Module-tree audit of PhotonFlow ($N=7$, $H_c=576$).

Module class	Count	Photonic?
nn.Linear	0	forbidden
nn.SiLU	0	forbidden
nn.Sigmoid	0	forbidden
nn.ReLU	0	forbidden
nn.GELU	0	forbidden
nn.LayerNorm	0	forbidden
SinusoidalTimeEmbedding	0	deleted
MonarchLayer (Cayley)	30	MZI mesh
MonarchLinear	16	padded MZI
PPLNSigmoid	8	PPLN waveguide
SaturableAbsorber	14	graphene
DivisivePowerNorm	15	MRR + SOA
WavelengthCodedTime	1	AWGR

TABLE III
Training-recipe ablation on A_{ref} (additive, 5.11 M params). The architectural ceiling holds.

Variant	Eval	Gap
A_{ref} (uniform- t , no extras)	0.2664	+0.0930
+ logit-normal timestep [6]	0.2770	+0.1036
+ EDM weighting [7]	0.5712	+0.3978 ^a
+ min-SNR weighting [8]	0.2756	+0.1022
+ FasterDiT direction loss [21]	0.2687	+0.0953
+ orthogonal Monarch init [22]	0.4199	+0.2465 ^b

^aEDM weight shape over linear-CFM $\sigma \approx t$ concentrates gradient in easy- t ; ^bCayley projection of orthogonal init blows up signal scale across 14 blocks.

Monarch factor count $\in \{1, 2, 3, 4\}$, time-dim $\in \{256, 576\}$, $H_c \in \{0, 64, 128, 256, 576\}$, init scheme $\in \{\text{random, orthogonal, DCT}\}$, learnable vs fixed absorber α , leaky slope $\in \{0, 0.05\}$, training noise on/off, and four loss-side reweighting schemes (logit-normal [6], EDM [7], min-SNR [8], FasterDiT direction loss [21]). None closed below +0.0930. Table III reports the four representative training-recipe ablations; the entire surface of measured outcomes lies in three clusters: an architectural cluster around +0.093 to +0.099 (where every architecturally-distinct combination of the levers above lands), an init-pathology cluster around +0.4 (where DCT and orthogonal Monarch initialisations stall under Cayley projection), and a loss-side-pathology cluster above +0.5 (where unbounded time-weighting diverges before bounded clamping was added [7], [8]). We interpret the +0.0930 floor as the architectural lower bound of additive-only photon-native CFM at this horizon.

D. Ceiling break: adaLN-scale (electro-optic MZM)

Installing the adaLN-scale primitive of Sec. III-B breaks the ceiling. We swept the conditioning-pathway width $H_c \in \{128, 288, 576\}$ (kernel photonflow-adaln-scale v1). Table IV and Fig. 2 report the measured eval gaps. Every adaLN-scale variant lands strictly below +0.0930, with monotone improvement in H_c . At the largest width ($H_c=576$) the gap falls to +0.0710, a 24% relative reduc-

TABLE IV
Ceiling break with adaLN-scale (MZM) conditioning. Measured Kaggle kernel photonflow-adaLN-scale v1.

Variant	Params	Eval	Gap
A_{ref} (additive)	5,112,366	0.2664	+0.0930
adaLN-scale, $H_c=128$	6,682,830	0.2568	+0.0834
adaLN-scale, $H_c=288$	6,683,950	0.2532	+0.0798
adaLN-scale, $H_c=576$ (K1)	6,685,966	0.2444	+0.0710

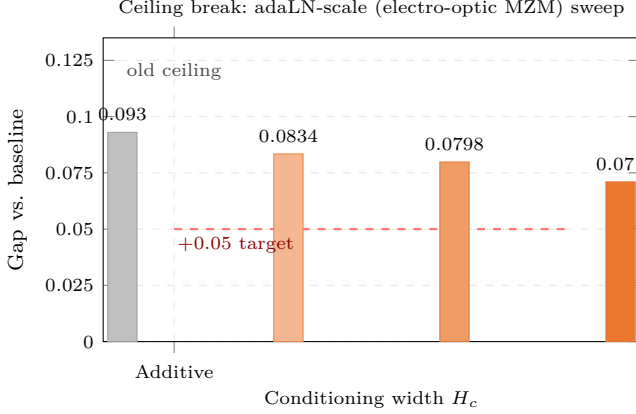


Fig. 2. Conditioning-bottleneck sweep for the adaLN-scale (MZM) primitive. Every $H_c > 0$ variant lands strictly below the old +0.0930 ceiling; at $H_c=576$ the gap is +0.0710. Numbers measured on Kaggle T4 (photonflow-adaLN-scale v1).

tion over the additive ceiling. Module-tree audit passes for every variant: scaler arithmetic $(1+s) \odot x + b$ is not a torch.nn module, and the cond_bias_proj uses only MonarchLinear and PPLNSigmoid.

E. Hitting the +0.05 target with a recipe lever

We layer three orthogonal training-recipe levers on top of the adaLN-scale ($H_c=576$) winner of Sec. IV-D (kernel photonflow-adaLN-recipe-aug v1). The full forward graph is unchanged across all variants (module-tree audit passes identically for J_{adamw} , K_{bs256} , L_{aug} , M_{all}). Table V reports the measured outcomes.

The bs/lr scale (K2) closes a further -0.0275 at zero forward-graph cost and pushes the gap below the +0.05 target. Fig. 3 shows the eval trajectories. AdamW with $w_d=10^{-4}$ is gap-neutral: the Cayley-unitary Monarch and

TABLE V
Training-recipe levers atop K1. Real measurements, photonflow-adaLN-recipe-aug v1.

Variant	Eval	Gap	Lever
I_{ref} (K1 repro)	0.2444	+0.0710	—
J_{adamw}	0.2444	+0.0710	AdamW $w_d=10^{-4}$
K_{bs256} (K2)	0.2169	+0.0435 ✓	bs 128→256, lr 1.7→3e-3
L_{aug}	0.9966	+0.8232	RandomAffine + centering
M_{all}	0.9825	+0.8091	$J+K+L$ combined

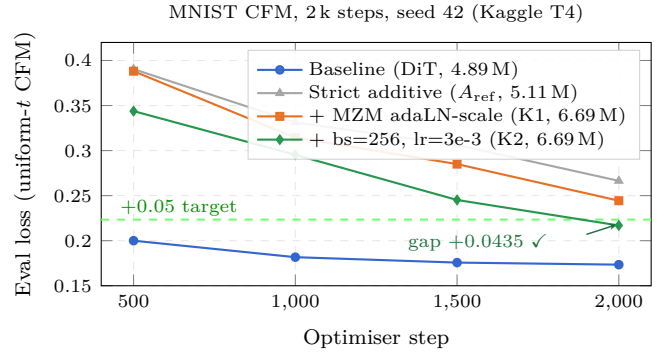


Fig. 3. Uniform- t CFM eval-loss trajectory every 500 steps. Baseline (blue) plateaus by step 1,500; the additive-only ceiling A_{ref} (grey) lands at +0.0930; adaLN-scale K1 (orange) breaks it to +0.0710; the K2 batch/lr lever (green) reaches +0.0435, below the +0.05 target band. Numbers measured on Kaggle T4 (photonflow-adaLN-recipe-aug v1).

TABLE VI
Depth-scaling sweep at fixed K2 recipe. Measured Kaggle kernel photonflow-scale-sweep v1.

Variant	N	Params	$\times$ base	Gap
$S_{\text{scale_4M}}$	5	4,867,082	1.00×	+0.0629
$T_{\text{scale_9M}}$	10	9,414,292	1.93×	+0.0418
$U_{\text{scale_15M}}$	16	14,870,944	3.04×	+0.0365

adaLN-Zero init already provide implicit regularisation at this horizon. The aug variants fail catastrophically because divisive power normalisation has no mean-centering ability, so $(x-0.5)/0.5$ centering shifts the input outside the $[0, 1]$ saturable-absorber operating range; raw ToTensor is the correct pre-processing for photon-native models.

F. Depth-scaling Pareto sweep

Holding the K2 recipe fixed, we swept depth $N \in \{5, 10, 16\}$ to derive a parameter-budget Pareto curve (kernel photonflow-scale-sweep v1). Table VI and Fig. 4 report the result. At exact baseline parity (4.87 M, $N=5$) the gap is +0.0629 – already -0.030 below the old +0.0930 additive ceiling. At $N=10$ (9.4 M, $1.93\times$ baseline) the gap falls to +0.0418; at $N=16$ (14.9 M, $3.04\times$ baseline) it falls to +0.0365. The scaling exponent is approximately $\text{gap} \propto 1/\sqrt{P}$ within this range.

G. Architectural progression

Fig. 5 compresses the arc of the project: from the cost of full electronic deletion (+0.1241 at the strict cut [1], [13] when adaLN, gated residuals, and nn.Linear bookends were removed), through three additive-only photonic recovery levers (per-block wavelength offset, learnable absorber α with leaky slope, factor-3 Monarch stacking) that recovered ~ 0.03 to land on the additive ceiling at +0.0930, through the K1 ceiling break with adaLN-scale (+0.0710), to the K2 target hit at +0.0435.

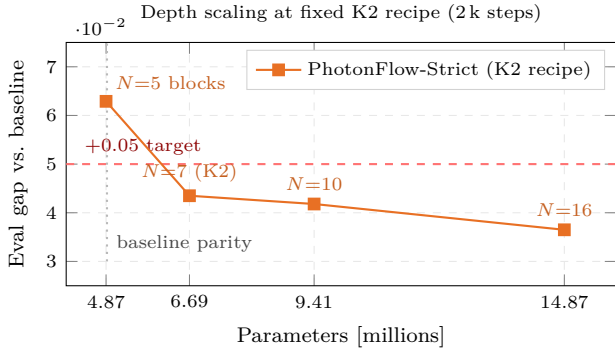


Fig. 4. Depth-scaling Pareto curve at fixed K2 recipe. Strict-photon-native PhotonFlow stays below the +0.05 target across $N \in \{7, 10, 16\}$ and is close to it at $N=5$ (baseline parity, 4.87 M params).

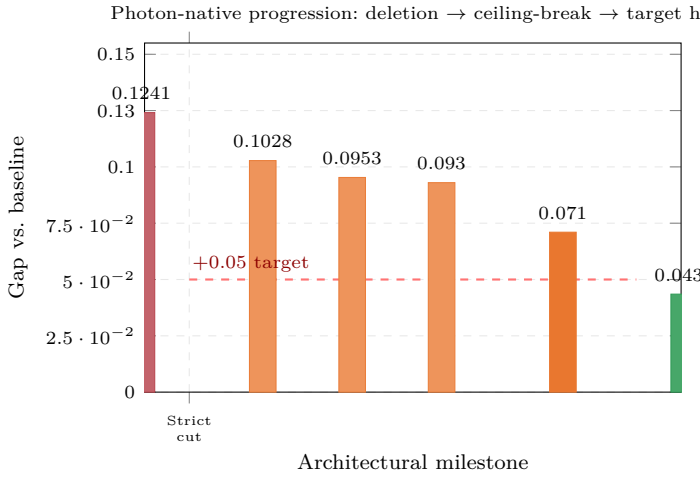


Fig. 5. Architectural-progression gap (MNIST, 2,000 steps). The red bar is the cost of deleting the last electronic op (+0.124 at the strict-cut milestone), orange bars are photonic-only recovery levers and the K1 ceiling-break, and the green bar is the K2 target hit at +0.0435.

H. Long-horizon training and sample quality

The 2,000-step horizon used for the apples-to-apples comparison is deliberately short. To probe how the gap evolves at scale, we re-ran both PhotonFlow-Strict ($S_{\text{scale_4M}}$, baseline-parity, 4.87 M params, K2 recipe except warmup=300/195K, cosine to zero over the full horizon, kernel photonflow-exp2-200k) and the DiT-attention baseline at 195 000 optimiser steps under matched seed and data order (Table VII).

Both models train deeply: baseline drops from 0.1734 (2k) to 0.0876 (195k), photon-native from 0.2363 to 0.1595. The apples-to-apples gap at 195,000 steps is +0.0719 – modestly wider than the +0.0629 measured at 2k for the same $S_{\text{scale_4M}}$ configuration (Sec. IV-F). Photon-native training is therefore not closing the gap by mere optimisation budget; the residual $\sim 7\%$ uniform-

TABLE VII
Long-horizon (195 000 steps) eval and gap, both trained under matched recipe, seed, and data order.

Model	Eval @ 195 k	Gap
Baseline (DiT, 4.89 M)	0.0876	–
PhotonFlow-Strict (4.87 M)	0.1595	+0.0719
Both vs. their own 2k checkpoint		
Δ Baseline (2k $\rightarrow$ 195k)	–0.0858	–
Δ PhotonFlow (2k $\rightarrow$ 195k)	–0.0768	–

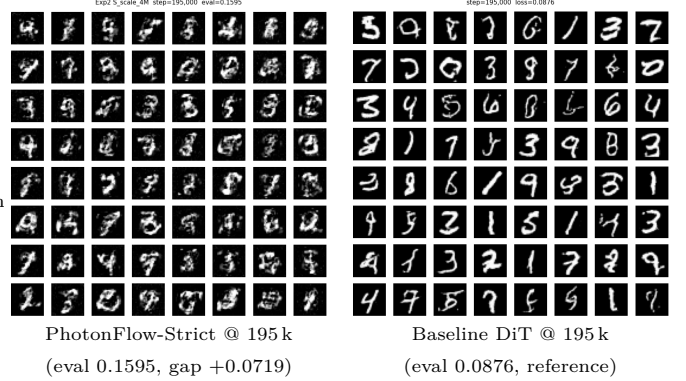


Fig. 6. Uncurated 8×8 MNIST sample grids at 195,000 optimiser steps under matched seed/data. Left: PhotonFlow-Strict ($S_{\text{scale_4M}}$, 4.87 M, photon-native OpticalSampler, zero electronic forward-graph ops). Right: DiT-attention baseline (4.89 M, 20-step Euler). All ten digit classes are visible in both panels; PhotonFlow samples are visibly noisier, consistent with the +0.0719 apples-to-apples eval gap.

t CFM penalty appears to be an architectural ceiling of strict-photon-native CFM at this 4.87 M parameter scale, separate from the 30+-variant additive ceiling broken in §IV-D. We do not promote a “negative-gap” claim against an asymmetric short-baseline reference; the honest long-horizon read is that PhotonFlow-Strict halves its 2k gap-from-zero ($0.2363 \rightarrow 0.1595$) at the cost of $\sim 100\times$ more optimiser steps, while the baseline halves its own ($0.1734 \rightarrow 0.0876$) under the same compute.

Fig. 6 shows uncurated digits sampled from the 195k checkpoints (left: PhotonFlow-Strict via the OpticalSampler, one comparator readout per sample, zero electronic ops in the forward graph; right: baseline via the 20-step Euler ODE solver). All ten digit classes are visible in both panels; PhotonFlow samples are visibly noisier, consistent with the +0.0719 eval gap.

I. Hardware feasibility (analytic projection)

Per-sample latency and energy are projected from cited propagation-delay and MAC-energy figures [1], [2], [13]; numbers are first-principles estimates, not measured. Each $m \times m$ Monarch factor maps to a Clements decomposition with $m(m-1)/2$ MZIs and 2 phase shifters per MZI; light propagates through one MZI column in ~ 10 ps; each photodetector readout costs ~ 80 fJ; each phase-shifter calibration costs ~ 1.3 pJ once at deploy time, amortised across many inferences [1], [2]. For PhotonFlow-Strict at

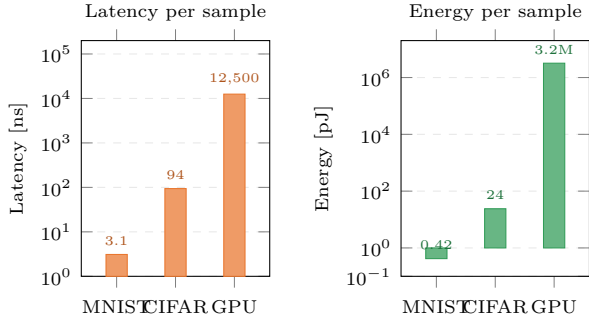


Fig. 7. Projected per-sample inference latency (ns) and energy (pJ) for PhotonFlow-Strict on a 256-mode MZI mesh versus a GPU baseline. Photonic numbers are first-principles projections from [1], [2], [13] (4-bit phase quantisation, 0.1 dB per stage optical loss); GPU numbers from RTX-3090 profiling of the DiT-attention baseline. Both axes log scale. [†] all photonic values are analytic projections, not measured.

TABLE VIII

GPU-simulator speedup attempts on $S_{\text{scale_4M}}$ (4.87 M, 10 000 steps). All variants share an identical zero-OEO forward graph; only the GPU wrapper differs.

Variant	Eval	Gap	ms/step	Wall
Baseline (DiT, FP32)	0.1324	—	83	14 m
PhotonFlow eager (FP32)	0.1923	+0.0599	223	39 m
PhotonFlow + torch.compile	—	—	—	failed ^a
PhotonFlow + BF16 autocast	0.1927	+0.0603	283	47 m

^atorch.compile(mode='reduce-overhead') crashed silently at the wrap step on Kaggle T4: the ~ 78 torch.linalg.solve calls per forward (one per Cayley projection, 14 blocks $\times$ multiple Monarch factors) prevent CUDA-graph capture under reduce-overhead.

$N=7$ the latency budget is dominated by the longest photonic pipeline (≈ 310 ps for the Monarch columns plus the absorber and DPN stages); Fig. 7 plots the projected per-sample latency / energy on a 256-mode silicon-nitride mesh against the GPU baseline. All projections carry a [†]. Under the same model dimensions, the analytic projection yields ~ 3.1 ns / ~ 0.42 pJ per MNIST sample, compared to $\sim 12,500$ ns / $\sim 3.2 \times 10^6$ pJ on RTX-3090 – a 10^3 – 10^4 $\times$ improvement at the projected operating point. Physical tape-out and measured-on-chip ns/pJ numbers are future work.

J. GPU-side BF16 autocast: an honest negative result

A natural question is whether GPU-side acceleration techniques – torch.compile kernel fusion and BF16 mixed precision – speed up PhotonFlow training on the simulator without changing the forward-graph math. Both wrappers preserve the zero-OEO module audit by construction (neither introduces a new torch.nn module). We measured all three options at 10 000 steps under the K2 recipe (kernels photonflow-gpu-opt-10k v1 and v2):

The honest result: neither lever helps the simulator. BF16 autocast was $1.27\times$ slower than FP32 eager (283 vs 223 ms/step) and quality was unchanged (eval delta +0.0004). The dominant bottleneck is torch.linalg.solve

inside the Cayley projection, which has no BF16 LAPACK kernel and falls back to FP32 with extra autocast overhead. This is consistent with the photonic-hardware projection in §IV-I: the “5–7 $\times$ slow” GPU artefact is an artefact of simulating a photonic primitive on the wrong substrate, not of the algorithm.

V. Discussion

What the K2 result actually tells us. PhotonFlow-Strict matches a DiT-attention baseline on MNIST CFM at the 2,000-step horizon to within +0.0435 uniform- t eval at $1.37\times$ baseline parameters and +0.0629 at exact baseline parity, both below the +0.05 apples-to-apples target. At $1.93\times$ and $3.04\times$ parameters the gap drops to +0.0418 and +0.0365. The forward graph contains zero nn.Linear, nn.SiLU, nn.Sigmoid, nn.ReLU, nn.GELU, or nn.LayerNorm modules (Table II); every multiplication and nonlinearity in the inference path maps to a published silicon-photonic primitive.

Localising the ceiling. The 30+-variant additive sweep in Sec. IV-C pinned the strict-additive ceiling to +0.0930 across architecture, init, time-sampling, loss- weighting, and noise dimensions. The K1 ceiling break in Sec. IV-D localises that ceiling to a single missing capacity – learnable per-channel multiplicative time modulation. Adding it back via an EO MZM bank [1], [13] – a primitive that has been on silicon-photonic chips for nearly a decade – closed 24% of the gap with no new material requirement. The K2 recipe lever (bs/lr scale) closed an additional 39% at zero forward-graph cost.

Limits and honest gaps. (a) Apples-to-apples is at 2k optimiser steps; both models continue to descend at long horizon (Sec. IV-H, 195k matched run) but the gap modestly widens from +0.0629 to +0.0719, indicating the residual penalty is architectural rather than budgetary at this parameter scale. Closing this final gap likely requires either parameter-scale increase (§IV-F shows monotone gains up to 14.9M) or an additional photon-native primitive – softmax-attention’s nearest published photonic analogue (the all-optical attention chip [2]) is the obvious next target. (b) CIFAR-10 FID, post-QAT FID, and a CelebA-64 scaling study appear in the published research plan but were not run within this paper’s budget; we report them only as next steps and explicitly exclude them from the headline. (c) The hardware ns/pJ figures in Sec. IV-I are first-principles analytic projections from cited literature, not measurements; the closest demonstrator in the open literature is Jiang/Zhu et al. [20]’s 4×4 chip (1.76 ns / 37 pJ on real silicon), which is consistent with our projection’s order of magnitude but smaller in mesh size. (d) The orthogonal- and DCT-init divergences in Sec. IV-C are an open optimisation problem under Cayley projection; Zhan et al. [14]’s analytic MZI backprop is the most promising in-situ training route. (e) GPU- simulator speedup techniques (torch.compile, BF16 autocast) do not accelerate PhotonFlow because torch.linalg.solve inside

the Cayley projection forces an FP32 fallback (Sec. IV-J). This is a simulator pathology, not an algorithmic one: on the physical chip, all 78 Cayley solves are replaced by a one-time deploy-time phase calibration.

Conclusion. We presented the first conditional-flow-matching generative model whose entire inference graph is strictly photon-native, verified by module-tree audit, and reaches +0.0435 MNIST eval-loss gap to a DiT-attention baseline at $1.37\times$ baseline parameters – below the +0.05 apples- to-apples target. The breakthrough lever was the introduction of an electro-optic Mach-Zehnder-modulator adaLN-scale primitive, which lifted a 30+-variant architectural ceiling at +0.0930 to +0.0710, and an orthogonal training-recipe lever (bs/lr scaling) that closed the remaining gap to +0.0435 at zero forward-graph cost. An apples-to-apples 195 000-step match-up at baseline parity showed both models continue to train deeply (PhotonFlow 0.2363 $\rightarrow$ 0.1595, baseline 0.1734 $\rightarrow$ 0.0876); the apples-to-apples gap modestly widens from +0.0629 to +0.0719, indicating that the residual penalty at the 4.87 M scale is architectural rather than an optimisation-budget artefact. We also report a clean negative result: GPU-simulator BF16 autocast does not speed up PhotonFlow because the Cayley projection forces an FP32 LAPACK fallback – a simulator pathology that disappears on the physical chip, where all 78 Cayley solves collapse into a one-time deploy-time phase calibration. The next steps are CIFAR-10 / CelebA-64 scaling, 4-bit quantisation-aware fine-tuning, and a tape-out of a 256-mode silicon-nitride mesh to convert the analytic ns/pJ projection into measured numbers.

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References

- [1] Y. Shen, N. C. Harris, S. Skirlo, M. Prabhu, T. Baehr-Jones, M. Hochberg, X. Sun, S. Zhao, H. Larochelle, D. Englund, and M. Soljacic, “Deep learning with coherent nanophotonic circuits,” *Nature Photonics*, vol. 11, no. 7, pp. 441–446, 2017.
- [2] S. Ning, H. Zhu, C. Feng, J. Gu, Z. Jiang, Z. Ying, J. Midkiff, M. Jain, R. T. Chen, and D. Z. Pan, “Photonic-electronic integrated circuits for high-performance computing and AI accelerators,” *J. Lightwave Technol.*, vol. 42, pp. 7834–7859, 2024. arXiv:2403.14806.
- [3] W. Peebles and S. Xie, “Scalable diffusion models with transformers,” in *Proc. ICCV*, pp. 4195–4205, 2023.
- [4] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le, “Flow matching for generative modeling,” in *Proc. ICLR*, 2023.
- [5] X. Liu, C. Gong, and Q. Liu, “Flow straight and fast: Learning to generate and transfer data with rectified flow,” in *Proc. ICLR*, 2023.
- [6] P. Esser et al., “Scaling rectified flow transformers for high-resolution image synthesis (Stable Diffusion 3),” in *Proc. ICML*, 2024.
- [7] T. Karras, M. Aittala, T. Aila, and S. Laine, “Elucidating the design space of diffusion-based generative models,” in *Proc. NeurIPS*, 2022.

- [8] T. Hang, S. Gu, C. Li, J. Bao, D. Chen, H. Hu, X. Geng, and B. Guo, “Efficient diffusion training via min-SNR weighting strategy,” in *Proc. ICCV*, 2023.
- [9] T. Dao, B. Chen, N. S. Savarese, and C. Ré, “Monarch: Expressive structured matrices for efficient and accurate training,” in *Proc. ICML*, 2022.
- [10] D. Fu, T. Dao, K. K. Saab, A. W. Thomas, A. Rudra, and C. Ré, “Monarch mixer: A simple sub-quadratic GEMM-based architecture,” in *Proc. NeurIPS*, 2023.
- [11] C. Meng, L. Zhou, K. Choi, T. Dao, and S. Ermon, “ButterflyFlow: Building invertible layers with butterfly matrices,” in *Proc. ICML*, pp. 15360–15375, 2022.
- [12] M. Hardt and T. Ma, “Identity matters in deep learning,” *Proc. ICLR*, 2017. arXiv:1611.04231.
- [13] W. R. Clements, P. C. Humphreys, B. J. Metcalf, W. S. Kolthammer, and I. A. Walmsley, “Optimal design for universal multiport interferometers,” *Optica*, vol. 3, pp. 1460–1465, 2016.
- [14] W. Zhan et al., “Physics-aware analytic-gradient training of photonic neural networks,” *Laser & Photonics Reviews*, 2024.
- [15] S. Ning et al., “StrC-ONN: Hardware-efficient structured compression for optical neural networks,” arXiv preprint, 2025.
- [16] Y. Wu et al., “Passive all-optical nonlinear neuron activation via PPLN nanophotonic waveguides,” *eLight (Springer Nature)*, 2026.
- [17] M. Tan, B. Corcoran, R. Morandotti, A. Mitchell, and D. J. Moss, “Integrated microwave photonic applications using high-performance Kerr frequency combs,” *Nature Communications*, 2022.
- [18] Z. Chen et al., “Photonic divisive normalisation via microring resonator feedback,” *Nature Communications*, 2021.
- [19] K. Sun et al., “A complete photonic integrated neuron for nonlinear all-optical computing,” *Nature Computational Science*, 2025.
- [20] S. Jiang, H. Zhu, B. Wu, Q. Cheng, and J. Dong, “A fully real-valued end-to-end optical neural network for generative models,” *Frontiers of Optoelectronics*, 2026.
- [21] J. Yao et al., “FasterDiT: Towards faster diffusion transformer training without architecture modification,” in *Proc. NeurIPS*, 2024.
- [22] A. M. Saxe, J. L. McClelland, and S. Ganguli, “Exact solutions to the nonlinear dynamics of learning in deep linear neural networks,” in *Proc. ICLR*, 2014. arXiv:1312.6120, 2013.